PROBLEM 41

Satellite Solar Panel: Composition of Rotations

Original statement · Problems.pdf, p. 29

Relative Kinematics

The angle θ describing the rotation of the solar panels of a reconnaissance satellite about the body-fixed axis x is an arbitrary function of time. The satellite spins about the z axis at the constant angular rate ω. Derive expressions for the absolute velocity and acceleration of point B in the figure relative to an inertial reference frame centered in A.

(Taken from Ginsberg, "Engineering Dynamics", Cambridge University Press, 2008.)

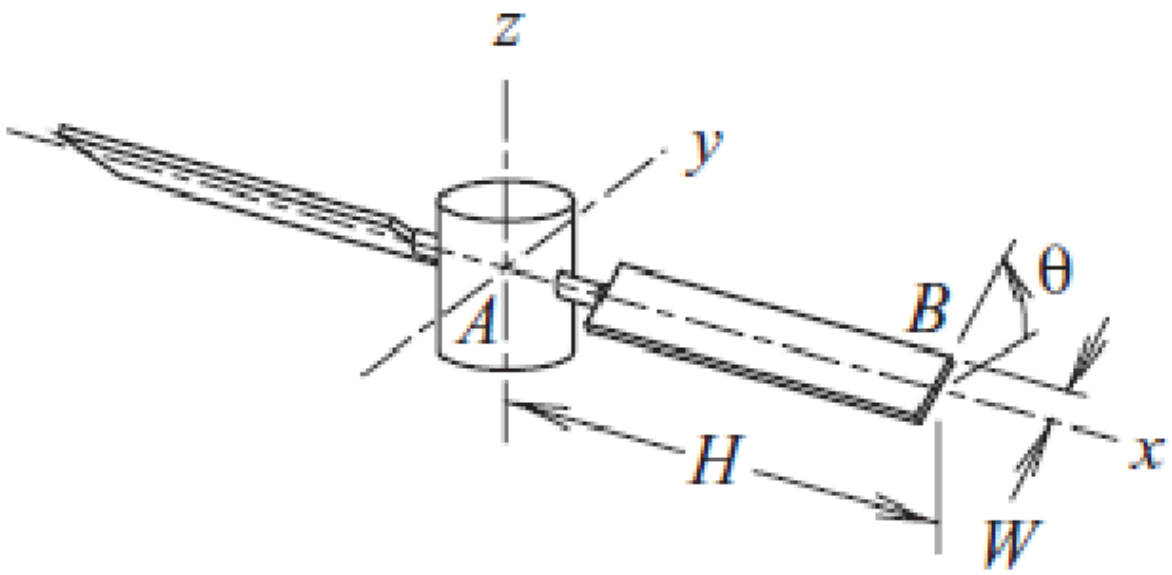


Figure: Satellite body spinning about z with a solar panel rotated by the angle θ about the body-fixed x axis, points A and B, dimensions H and W (Problems.pdf, p. 29).

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a satellite body spins about its z axis at constant rate ω; a solar panel turns about the body-fixed x axis through the prescribed angle $\theta(t)$. Point B is a corner of the panel.
- **Why relative motion.** The motion of B is the superposition of two simple motions: in the body frame S_1, B just describes a circle of radius W about the x axis; and S_1 itself just spins about z. Notes Eqs. 3.26–3.27 combine the two without writing the inertial position explicitly. A second route uses the panel frame S_2, in which B is fixed, and the composition of angular velocities (Notes Eqs. 3.13–3.15).
- **Hypotheses:**
  1. The frame centred at A with non-rotating axes is inertial, so the centre A of the satellite is fixed: $\mathbf{v}_0^A = \mathbf{a}_0^A = \mathbf{0}$.
  2. Body and panel are rigid; ω is constant; $\theta(t)$ is an arbitrary (known) function.
  3. **Geometry (read from the figure):** B is at distance H from A measured along x, and at distance W from the x axis; θ is measured about $+x$ from the body axis y, so for $\theta = 0$ the segment from the hinge axis to B points along $+y$.
- **Inertial frame:** $S_0 = \{A; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$.
- **Body frame:** $S_1 = \{A; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ (the axes x, y, z of the figure), basis $\mathcal{B}_1$, turning with the satellite about $\mathbf{k}_0 = \mathbf{k}_1$. It coincides with S_0 at $t = 0$.
- **Panel frame:** $S_2 = \{C; \mathbf{i}_2, \mathbf{j}_2, \mathbf{k}_2\}$, basis $\mathcal{B}_2$, attached to the panel, where $C = A + H\,\mathbf{i}_1$ is the point of the hinge axis at the abscissa of B; $\mathbf{i}_2 = \mathbf{i}_1$ is the hinge axis and $\mathbf{j}_2$ points from C to B.
- **Degrees of freedom:** none to determine; the motion is prescribed by ωt and $\theta(t)$ (kinematics only).

Symbol	Meaning	SI unit
H	axial distance from A to B (along x)	m
W	distance from B to the hinge axis x	m
ω	constant spin rate of the satellite about z	rad/s
$\theta(t)$	panel angle about x	rad

Change of Basis and Position Vector

First rotation: the body frame S_1 about $\mathbf{k}_0$

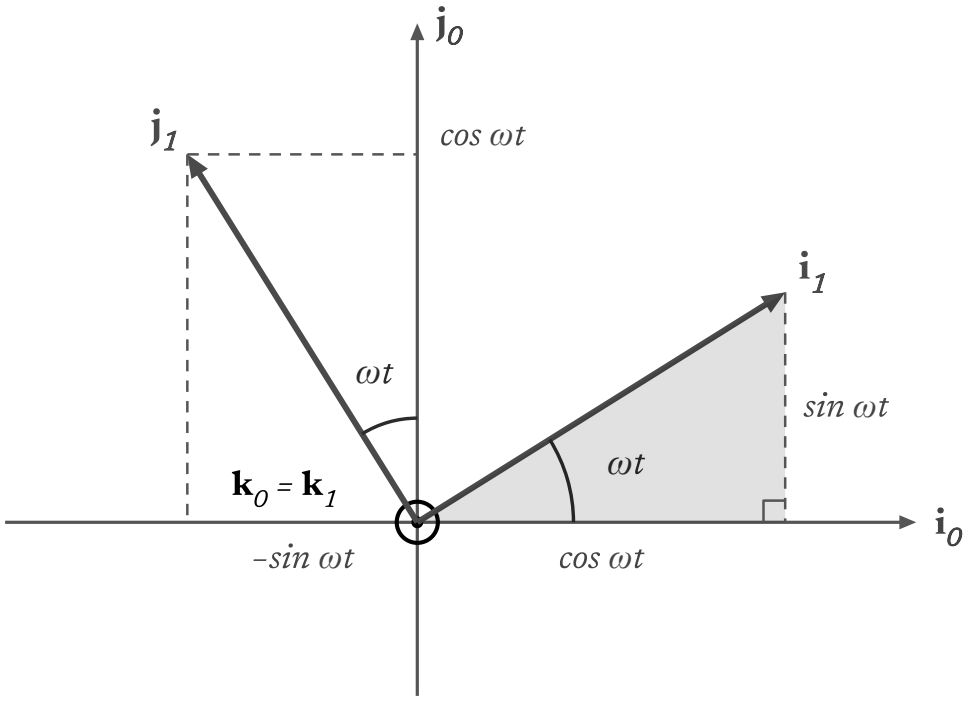


Figure 1. First rotation, seen from $+z$ with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$): $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by ωt. $\mathbf{i}_1$ has components $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $\mathbf{j}_0$; the tip of $\mathbf{j}_1$ moves to the side of $-\mathbf{i}_0$, hence its component $-\sin\omega t$.

The spin is about $\mathbf{k}_0$, so we look down that axis, from $+z$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule fixes $\mathbf{j}_0$: $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$ means that the fingers turning from $\mathbf{i}_0$ to $\mathbf{j}_0$ must curl counter-clockwise for the thumb to point at the reader, so $\mathbf{j}_0$ points up the page. A positive spin $\omega\,\mathbf{k}_0$ turns the body counter-clockwise in this view, and after a time t the body axes have turned the angle ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded right triangle. The leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$; the leg opposite to ωt lies along $+\mathbf{j}_0$ and measures $\sin\omega t$. So $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up) and turns the same angle counter-clockwise, which moves its tip to the left, to the side of $-\mathbf{i}_0$. Its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $\mathbf{i}_0$ (opposite) is $\sin\omega t$ pointing to $-\mathbf{i}_0$: $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$. **The minus sign comes from the tip**

of $\mathbf{j}_1$ moving towards $-\mathbf{i}_0$.

- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The columns of the matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = 1\cdot(\cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t) = \cos^2\omega t + \sin^2\omega t = 1$$

(expansion along the third row). The columns have unit norm, $\cos^2\omega t + \sin^2\omega t = 1$, and the first two are orthogonal, $\cos\omega t\,(-\sin\omega t) + \sin\omega t\cos\omega t = 0$, so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1. $\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1$ and, ω and $\mathbf{k}_1 = \mathbf{k}_0$ being constant, $\boldsymbol{\alpha}_{10} = \mathbf{0}$. Check with Poisson's formula: by the chain rule, $\left.\frac{d\mathbf{i}_1}{dt}\right|_0 = -\omega\sin\omega t\,\mathbf{i}_0 + \omega\cos\omega t\,\mathbf{j}_0 = \omega\,\mathbf{j}_1$, and indeed $\boldsymbol{\omega}_{10}\times\mathbf{i}_1 = \omega\,\mathbf{k}_1\times\mathbf{i}_1 = \omega\,\mathbf{j}_1$.

Second rotation: the panel frame S_2 about the hinge $\mathbf{i}_1$

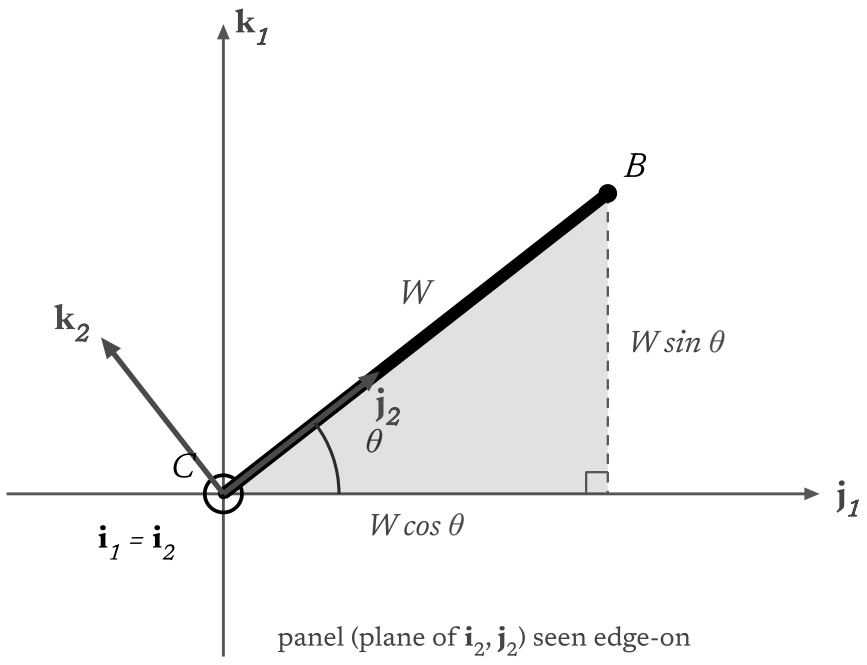


Figure 2. Second rotation, seen along the hinge with $\mathbf{i}_1 = \mathbf{i}_2$ out of the paper ($\odot$), so $\mathbf{j}_1$ points right and $\mathbf{k}_1 = \mathbf{i}_1\times\mathbf{j}_1$ up. The panel turns θ counter-clockwise: $CB = W\,\mathbf{j}_2$ is the hypotenuse of the shaded triangle, with legs $W\cos\theta$ along $\mathbf{j}_1$ and $W\sin\theta$ along $\mathbf{k}_1$.

The panel turns about the hinge axis $\mathbf{i}_1 = \mathbf{i}_2$, so we look along that axis with $\mathbf{i}_1$ out of the paper ($\odot$). With $\mathbf{j}_1$ to the right, $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$: the fingers go from $\mathbf{i}_1$ (towards the reader) to $\mathbf{j}_1$ (right) and the thumb points up the page, so $\mathbf{k}_1$ points up. The angle θ is measured about $+\mathbf{i}_1$, which is counter-clockwise in this view, from $\mathbf{j}_1$:

- $\mathbf{j}_2$ has the leg $\cos\theta$ along $\mathbf{j}_1$ (adjacent) and the leg $\sin\theta$ along $+\mathbf{k}_1$ (opposite): $\mathbf{j}_2 = \cos\theta\,\mathbf{j}_1 + \sin\theta\,\mathbf{k}_1$.
- $\mathbf{k}_2$ starts on $\mathbf{k}_1$ and its tip moves to the side of $-\mathbf{j}_1$: $\mathbf{k}_2 = -\sin\theta\,\mathbf{j}_1 + \cos\theta\,\mathbf{k}_1$.
- $\mathbf{i}_2 = \mathbf{i}_1$.

$$[{}_1R_2] = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}, \qquad \det[{}_1R_2] = 1 \cdot \left(\cos^2\theta + \sin^2\theta\right) = 1, \qquad \boldsymbol{\omega}_{21} = \dot{\theta}\,\mathbf{i}_1$$

The columns are orthonormal (norms $\cos^2\theta + \sin^2\theta = 1$, scalar product of the last two $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$), so $[{}_1R_2][{}_1R_2]^T = I$; hence $[{}_0R_2] = [{}_0R_1][{}_1R_2]$ is also a proper rotation, with $\det[{}_0R_2] = 1 \cdot 1 = 1$ (Notes Eq. 1.16).

Position vector of B

To go from A to B we pass through the hinge point C, because the distance from the hinge axis is the one that stays constant:

$$\mathbf{r}_1^B = \overrightarrow{AC} + \overrightarrow{CB}$$

- $\overrightarrow{AC} = H\,\mathbf{i}_1$: C is on the hinge axis, at the abscissa H of B.
- $\overrightarrow{CB} = W\,\mathbf{j}_2$ is fixed in S_2. In the shaded triangle of the hinge view the hypotenuse is $|CB| = W$, the leg adjacent to θ is $W\cos\theta$ along $\mathbf{j}_1$ and the leg opposite to θ is $W\sin\theta$ along $\mathbf{k}_1$; this is the second column of $[{}_1R_2]$ times W.

$$\mathbf{r}_1^B = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$$

Checks: $\theta = 0$ gives $H\,\mathbf{i}_1 + W\,\mathbf{j}_1$ (panel along $+y$, as in the reading of the figure); $\theta = \pi/2$ gives $H\,\mathbf{i}_1 + W\,\mathbf{k}_1$ (panel turned up, along $+z$).

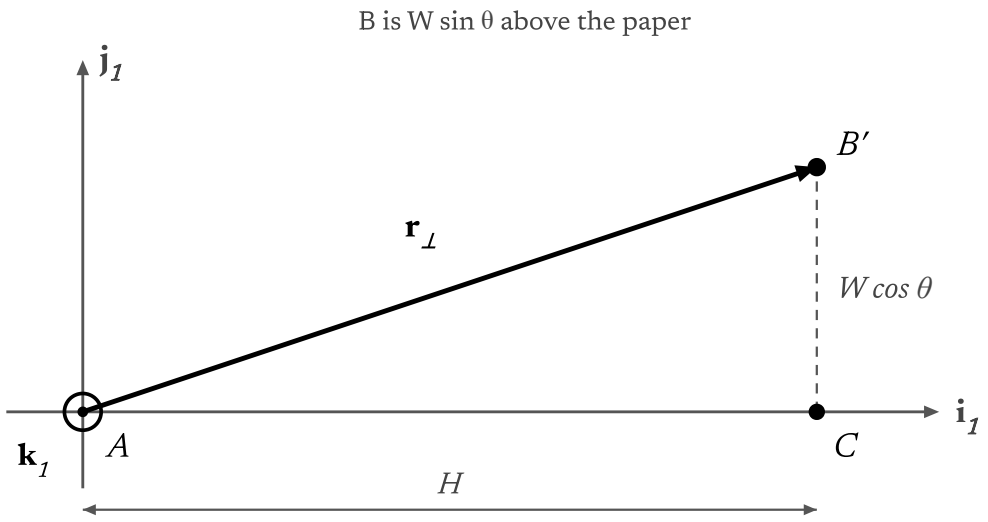


Figure 3. Body plane seen from $+z$ ($\mathbf{k}_1$ out of the paper): C is at H along $\mathbf{i}_1$ and the corner B projects onto the point B' at $W\cos\theta$ along $\mathbf{j}_1$ (B itself is $W\sin\theta$ above the paper). $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$ is the distance vector from the spin axis, the one that enters the centripetal term.

Distance to the spin axis. The spin is about $\mathbf{k}_1$, so the part of $\mathbf{r}_1^B$ that matters for the rotation is the one perpendicular to $\mathbf{k}_1$: dropping the component $W\sin\theta\,\mathbf{k}_1$ (the height of B above the body plane), $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$, the vector from the axis to the projection B' in the figure above. It reappears in the centripetal term.

Kinematics

1. Relative motion of B in the body frame S_1

Pedagogical Justification: the body frame rotates at constant rate with fixed origin, so Notes Eq. 3.27 keeps only three terms; the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant.

Only the components of $\mathbf{r}_1^B$ change. By the chain rule, $\frac{d}{dt}(W\cos\theta) = -W\sin\theta\,\dot\theta$ and $\frac{d}{dt}(W\sin\theta) = W\cos\theta\,\dot\theta$ (H is constant):

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -W\dot\theta\sin\theta\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1$$

Differentiating again with the product rule, $\frac{d}{dt}(\dot\theta\sin\theta) = \ddot\theta\sin\theta + \dot\theta\,\dot\theta\cos\theta$ and $\frac{d}{dt}(\dot\theta\cos\theta) = \ddot\theta\cos\theta - \dot\theta\,\dot\theta\sin\theta$:

$$\mathbf{a}_1^B = \left.\frac{d\mathbf{v}_1^B}{dt}\right|_1 = -W\left(\ddot\theta\sin\theta + \dot\theta^2\cos\theta\right)\mathbf{j}_1 + W\left(\ddot\theta\cos\theta - \dot\theta^2\sin\theta\right)\mathbf{k}_1$$

(circular motion of radius W about the x axis: tangential part $W\ddot\theta\,\mathbf{k}_2$ and normal part $-W\dot\theta^2\,\mathbf{j}_2$).

2. Absolute velocity of B

> **Pedagogical Justification:** Notes Eq. 3.26 (Slide 24), $\mathbf{v}_0^B = \mathbf{v}_0^A + \mathbf{v}_1^B + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$, with $\mathbf{v}_0^A = \mathbf{0}$.

Cross products: $\mathbf{k}_1\times\mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1\times\mathbf{j}_1 = -\mathbf{i}_1$, $\mathbf{k}_1\times\mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B = \omega\mathbf{k}_1\times(H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = H\omega\,\mathbf{j}_1 - W\omega\cos\theta\,\mathbf{i}_1$$

(only $\mathbf{r}_\perp$ contributes). Adding $\mathbf{v}_1^B$ component by component:

$$\boxed{\mathbf{v}_0^B = -W\omega\cos\theta\,\mathbf{i}_1 + \left(H\omega - W\dot\theta\sin\theta\right)\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1}$$

The components are given in the body basis $\mathcal{B}_1$; in $\mathcal{B}_0$ they follow from $\{\mathbf{v}_0^B\}_0 = [{}_0R_1]\{\mathbf{v}_0^B\}_1$.

3. Absolute acceleration of B

> **Pedagogical Justification:** Notes Eq. 3.27 (Slide 25) with $\mathbf{a}_0^A = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$: $\mathbf{a}_0^B = \mathbf{a}_1^B + \boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B) + 2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B$.

Centripetal term (reusing $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$ from step 2):

$$\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B) = \omega\mathbf{k}_1\times(-W\omega\cos\theta\,\mathbf{i}_1 + H\omega\,\mathbf{j}_1) = -W\omega^2\cos\theta\,\mathbf{j}_1 - H\omega^2\,\mathbf{i}_1$$

i.e. $-\omega^2\mathbf{r}_\perp$ with $\mathbf{r}_\perp = H\,\mathbf{i}_1 + W\cos\theta\,\mathbf{j}_1$, the vector of the body-plane figure.

Coriolis term:

$$2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2\omega\mathbf{k}_1\times\left(-W\dot\theta\sin\theta\,\mathbf{j}_1 + W\dot\theta\cos\theta\,\mathbf{k}_1\right) = -2W\omega\dot\theta\sin\theta\,(\mathbf{k}_1\times\mathbf{j}_1) = 2W\omega\dot\theta\sin\theta\,\mathbf{i}_1$$

Sum, collecting the $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{k}_1$ components of $\mathbf{a}_1^B$, the centripetal term and the Coriolis term:

$$\boxed{\mathbf{a}_0^B = \left(2W\omega\dot\theta\sin\theta - H\omega^2\right)\mathbf{i}_1 - \left(W\ddot\theta\sin\theta + W\dot\theta^2\cos\theta + W\omega^2\cos\theta\right)\mathbf{j}_1 + \left(W\ddot\theta\cos\theta - W\dot\theta^2\sin\theta\right)\mathbf{k}_1}$$

4. Second route: panel frame and composition of angular velocities

Pedagogical Justification: B is fixed in S_2, so $\mathbf{v}_2^B = \mathbf{a}_2^B = \mathbf{0}$ and Notes Eqs. 3.26–3.27 applied to S_2 (origin C) need only $\boldsymbol{\omega}_{20}$, $\boldsymbol{\alpha}_{20}$ and the motion of C. Angular velocities add (Notes Eqs. 3.13–3.15, Slide 19); angular accelerations do not simply add.

$$\boldsymbol{\omega}_{20} = \boldsymbol{\omega}_{21} + \boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1$$

Its derivative in S_0 by the Coriolis theorem (Notes Eq. 3.8), with $\frac{d}{dt}\big|_1\left(\dot\theta\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) = \ddot\theta\,\mathbf{i}_1$ (ω constant):

$$\boldsymbol{\alpha}_{20} = \left.\frac{d\boldsymbol{\omega}_{20}}{dt}\right|_1 + \boldsymbol{\omega}_{10}\times\boldsymbol{\omega}_{20} = \ddot\theta\,\mathbf{i}_1 + \omega\mathbf{k}_1\times\dot\theta\,\mathbf{i}_1 = \ddot\theta\,\mathbf{i}_1 + \omega\dot\theta\,\mathbf{j}_1$$

Motion of C (fixed in S_1 at $H\mathbf{i}_1$): $\mathbf{v}_0^C = \omega\mathbf{k}_1\times H\mathbf{i}_1 = H\omega\,\mathbf{j}_1$, $\mathbf{a}_0^C = \omega\mathbf{k}_1\times H\omega\,\mathbf{j}_1 = -H\omega^2\,\mathbf{i}_1$. With $\mathbf{r}_2^B = \overrightarrow{CB} = W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1$:

$$\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B = \dot\theta W\cos\theta\,\mathbf{k}_1 - \dot\theta W\sin\theta\,\mathbf{j}_1 - \omega W\cos\theta\,\mathbf{i}_1$$

so $\mathbf{v}_0^B = \mathbf{v}_0^C + \boldsymbol{\omega}_{20}\times\mathbf{r}_2^B$ reproduces the boxed velocity. For the acceleration, $\mathbf{a}_0^B = \mathbf{a}_0^C + \boldsymbol{\alpha}_{20}\times\mathbf{r}_2^B + \boldsymbol{\omega}_{20}\times(\boldsymbol{\omega}_{20}\times\mathbf{r}_2^B)$:

$$\boldsymbol{\alpha}_{20} \times \mathbf{r}_2^B = \left(\ddot{\theta}\,\mathbf{i}_1 + \omega\dot{\theta}\,\mathbf{j}_1\right) \times (W\cos\theta\,\mathbf{j}_1 + W\sin\theta\,\mathbf{k}_1) = W\ddot{\theta}\cos\theta\,\mathbf{k}_1 - W\ddot{\theta}\sin\theta\,\mathbf{j}_1 + W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{20} \times \left(\boldsymbol{\omega}_{20} \times \mathbf{r}_2^B\right) = \left(\dot{\theta}\,\mathbf{i}_1 + \omega\,\mathbf{k}_1\right) \times \left(-\omega W\cos\theta\,\mathbf{i}_1 - \dot{\theta}W\sin\theta\,\mathbf{j}_1 + \dot{\theta}W\cos\theta\,\mathbf{k}_1\right)$$

$$= -\dot{\theta}^2 W\sin\theta\,\mathbf{k}_1 - \dot{\theta}^2 W\cos\theta\,\mathbf{j}_1 - \omega^2 W\cos\theta\,\mathbf{j}_1 + \omega\dot{\theta}W\sin\theta\,\mathbf{i}_1$$

(using $\mathbf{i}_1 \times \mathbf{j}_1 = \mathbf{k}_1$, $\mathbf{i}_1 \times \mathbf{k}_1 = -\mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Adding $\mathbf{a}_0^C = -H\omega^2\,\mathbf{i}_1$: the $\mathbf{i}_1$ component is $2W\omega\dot{\theta}\sin\theta - H\omega^2$, the $\mathbf{j}_1$ component $-W(\ddot{\theta}\sin\theta + \dot{\theta}^2\cos\theta + \omega^2\cos\theta)$, the $\mathbf{k}_1$ component $W(\ddot{\theta}\cos\theta - \dot{\theta}^2\sin\theta)$: the same boxed result. Note that the Coriolis term of route 1, $2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$, appears in route 2 split in two equal halves, one from $\boldsymbol{\alpha}_{20}$ (the term $\omega\dot{\theta}\,\mathbf{j}_1$ that makes angular accelerations not add) and one from the double cross product.

Answers and Checks

Absolute velocity and acceleration of B

Both routes give, in the body basis $\mathcal{B}_1$ (to be multiplied by $[{}_0R_1]$ for the components in $\mathcal{B}_0$):

$$\mathbf{v}_0^B = -W\omega\cos\theta\,\mathbf{i}_1 + \left(H\omega - W\dot{\theta}\sin\theta\right)\mathbf{j}_1 + W\dot{\theta}\cos\theta\,\mathbf{k}_1$$

$$\mathbf{a}_0^B = \left(2W\omega\dot{\theta}\sin\theta - H\omega^2\right)\mathbf{i}_1 - \left(W\ddot{\theta}\sin\theta + W\dot{\theta}^2\cos\theta + W\omega^2\cos\theta\right)\mathbf{j}_1 + \left(W\ddot{\theta}\cos\theta - W\dot{\theta}^2\sin\theta\right)\mathbf{k}_1$$

No forces or masses are given, so the problem has no dynamics to write.

Checks

- **Reading of the terms.** $-H\omega^2\mathbf{i}_1 - W\omega^2\cos\theta\,\mathbf{j}_1 = -\omega^2\mathbf{r}_\perp$ is the centripetal acceleration of the spin; the $W\ddot{\theta}$ and $W\dot{\theta}^2$ terms are the tangential and normal accelerations of the panel rotation; $2W\omega\dot{\theta}\sin\theta\,\mathbf{i}_1$ is the Coriolis coupling of the two rotations, present only

when the panel moves ($\dot\theta \neq 0$) with a velocity component normal to the spin axis.

- **Limits:** $\dot\theta = 0$ (panel locked) gives the rigid-body motion $\mathbf{v}_0^B = \boldsymbol{\omega}_{10} \times \mathbf{r}_1^B$, $\mathbf{a}_0^B = -\omega^2 \mathbf{r}_\perp$; $\omega = 0$ (no spin) leaves the circular motion of radius W about the hinge.
- **Orthogonality:** $\mathbf{j}_2 \cdot \mathbf{k}_2 = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$ and $\mathbf{i}_1 \cdot \mathbf{j}_1 = -\cos\omega t \sin\omega t + \sin\omega t \cos\omega t = 0$.
- **Dimensions:** every velocity component is of the form $[W\omega] = [H\omega] = [W\dot\theta] = \mathrm{m\,s^{-1}}$; every acceleration component $[W\omega\dot\theta] = [H\omega^2] = [W\ddot\theta] = [W\dot\theta^2] = \mathrm{m\,s^{-2}}$.
- **Verification (sympy):** $[{}_0R_1]$, $[{}_1R_2]$, $[{}_0R_2]$ are proper rotations; the inertial first and second derivatives of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equal the Notes Eq. 3.26 and 3.27 expressions; both routes give exactly the official key.
- **Check against the official key:** the key of `Problems.pdf` (p. 76) for Problem 41 (absolute velocity and acceleration of B) is reproduced exactly by both routes, as the sympy verification above records; no discrepancy.

Source: Problems.pdf (PDF page 29); key on PDF page 76